AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Assignment

Assignment Title:	Inventory Management System-Selenium Testing			
Assignment No:	01		Date of Submission:	17 November 2026
Course Title:	SOFTWARE QUALITY AND TESTING			
Course Code:	CSC3112		Section:	B
Semester:	Fall	2026-27	Course Teacher:	DR. ABHIJIT BHOWMIK

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FACULTY COMMENTS	**Marks Obtained**	
	Total Marks	

1. Python Installation

Description:

Python is the programming language used to write the Selenium test scripts of this project. It was downloaded from the official Python website, installed on a Windows computer, and then verified from the Command Prompt.

The Downloads page of python.org was opened in the web browser. The “Download Python install manager” button was used to get the Windows installer.

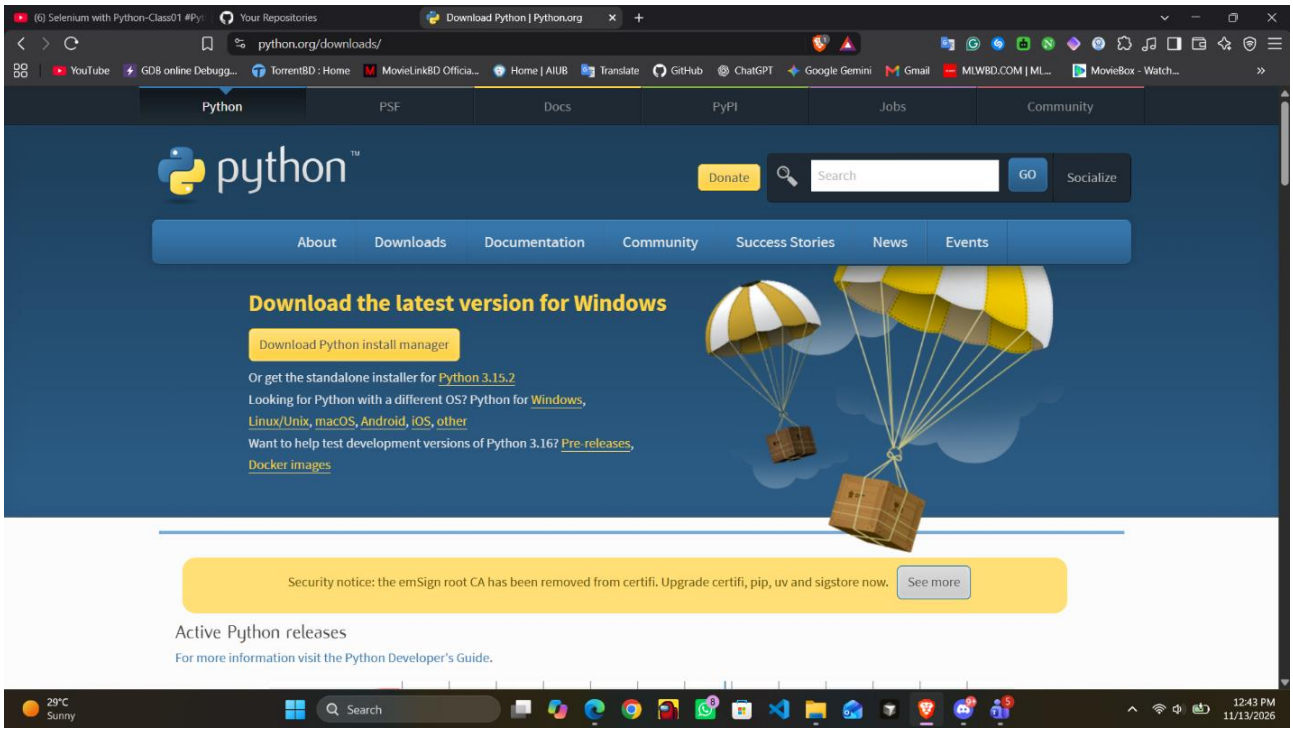


Fig. 01: Python download page on python.org

The installer file started downloading. The download window shows the file size, the amount already downloaded, the download speed and the remaining time.

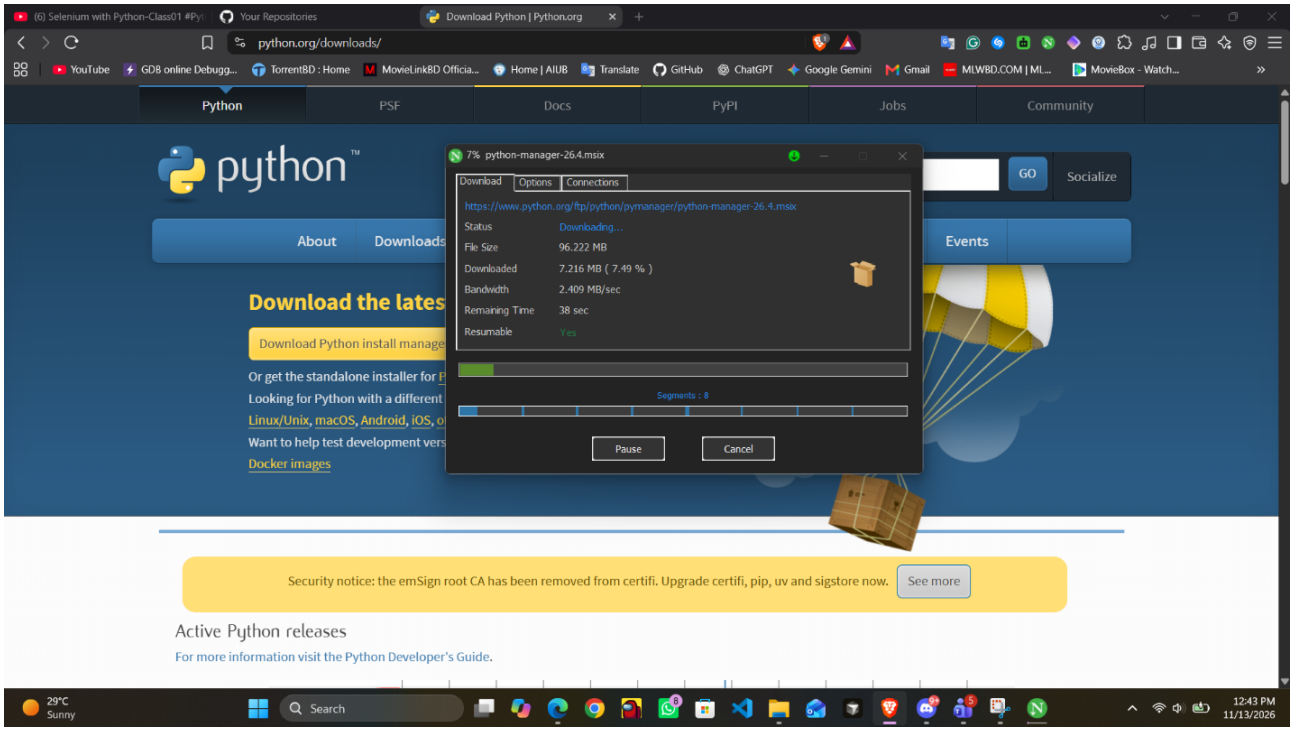


Fig. 02: Python installer download in progress

The downloaded installer was run. The Setup Progress window shows the Python 3.14.0 (64-bit) development libraries being installed.

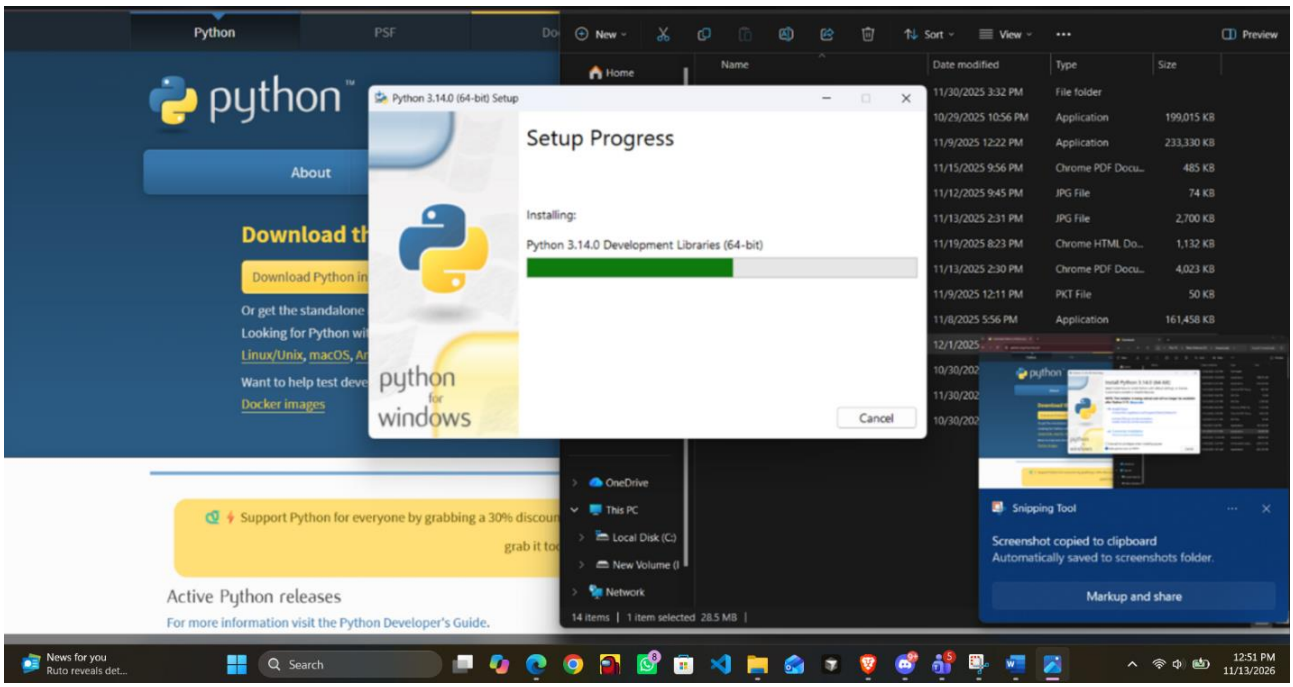


Fig. 03: Python setup in progress

When the installation finished, the setup wizard displayed the message “Setup was successful”, which confirms that Python was installed without any error.

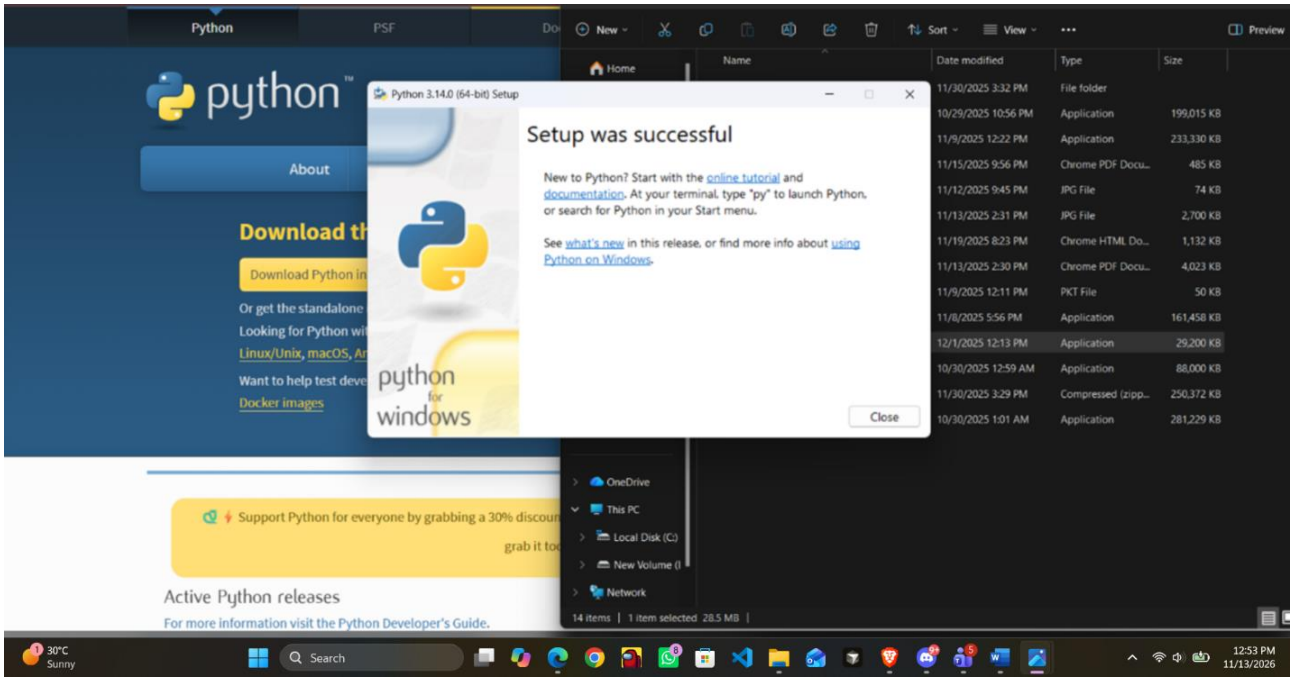


Fig. 04: Python setup completed successfully

The command python --version was executed in the Command Prompt. It returned the Python version, which confirms that Python is installed and can be used from the command line.

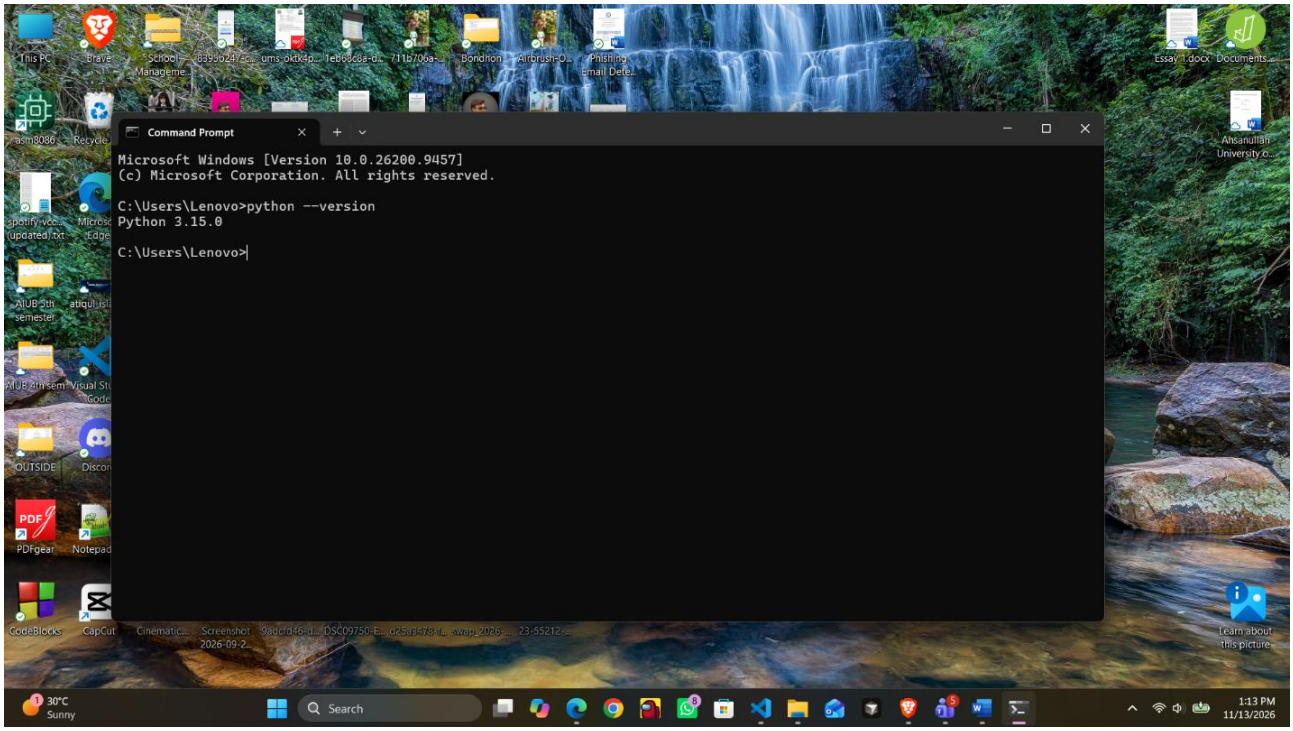


Fig. 05: Python version check in Command Prompt

2. Selenium Installation

Description:

Selenium is the automation library used to control the web browser during testing. It was installed with pip, the package manager of Python, and its setup was checked inside Visual Studio Code.

The Python version was checked first, and then the command pip install selenium was executed. Pip collected Selenium 4.52.0 together with the packages it depends on, such as trio, certifi, urllib3 and websocket-client.

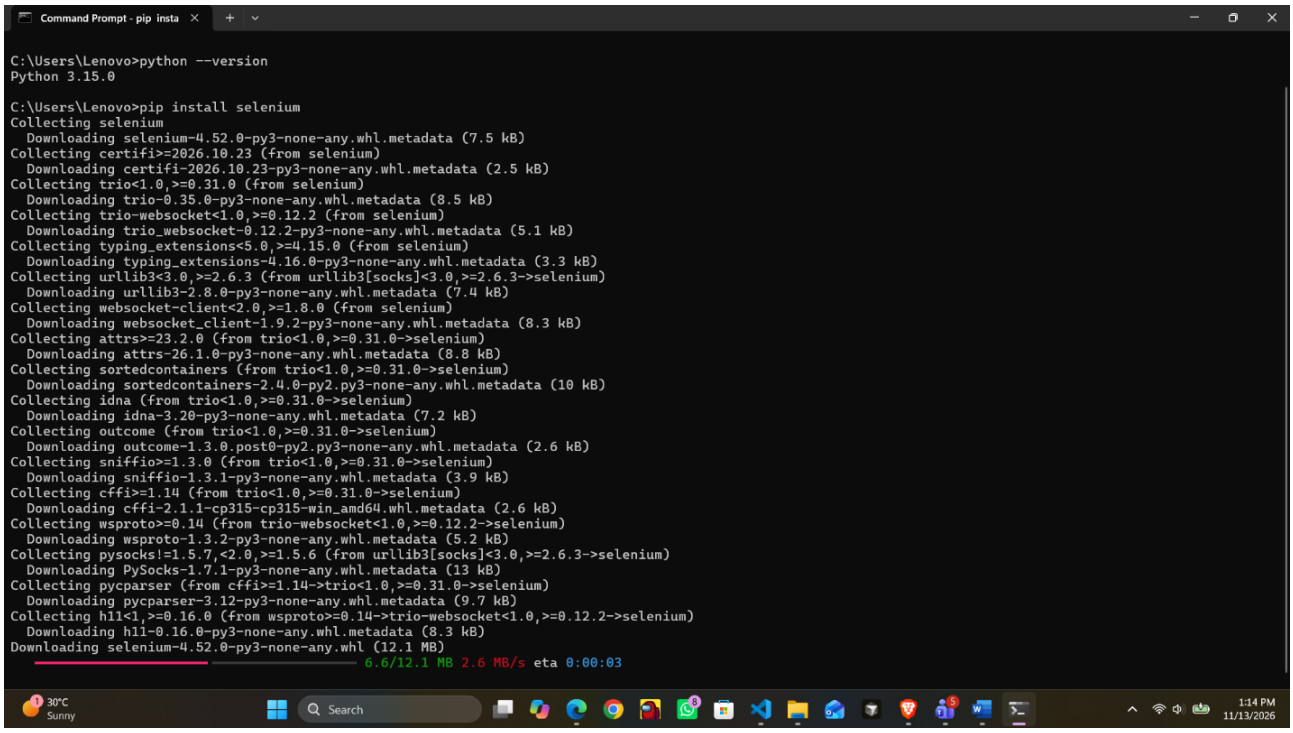


Fig. 06: Installing Selenium with pip

Pip finished with the message “Successfully installed”, listing Selenium and its dependencies. Pip also reported that a newer version was available, so it was upgraded

with the command python -m pip install --upgrade pip. The warnings about the Scripts folder not being in PATH do not affect the test scripts.

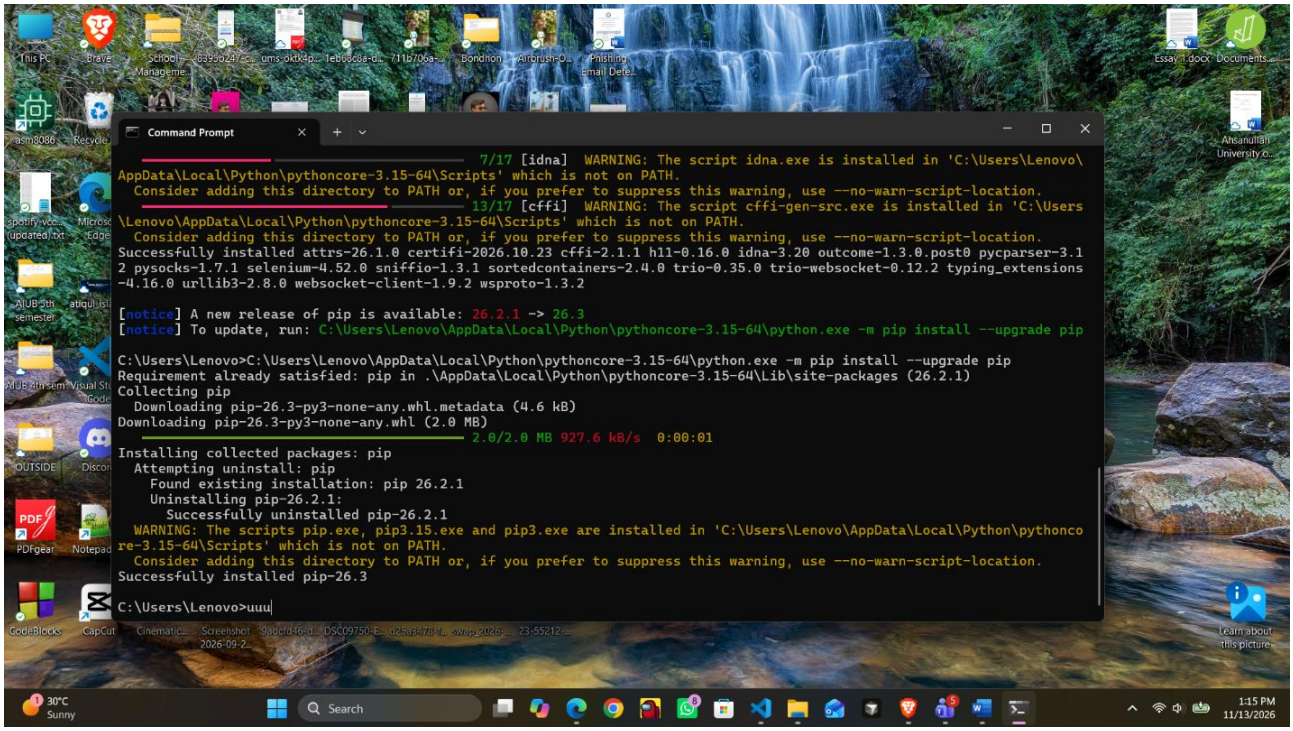


Fig. 07: Selenium installation completed

The project InventoryManagement-Selenium was opened in Visual Studio Code. The script imports webdriver, By, WebDriverWait and expected_conditions from Selenium and creates a Chrome driver, which confirms that Selenium is ready to use.

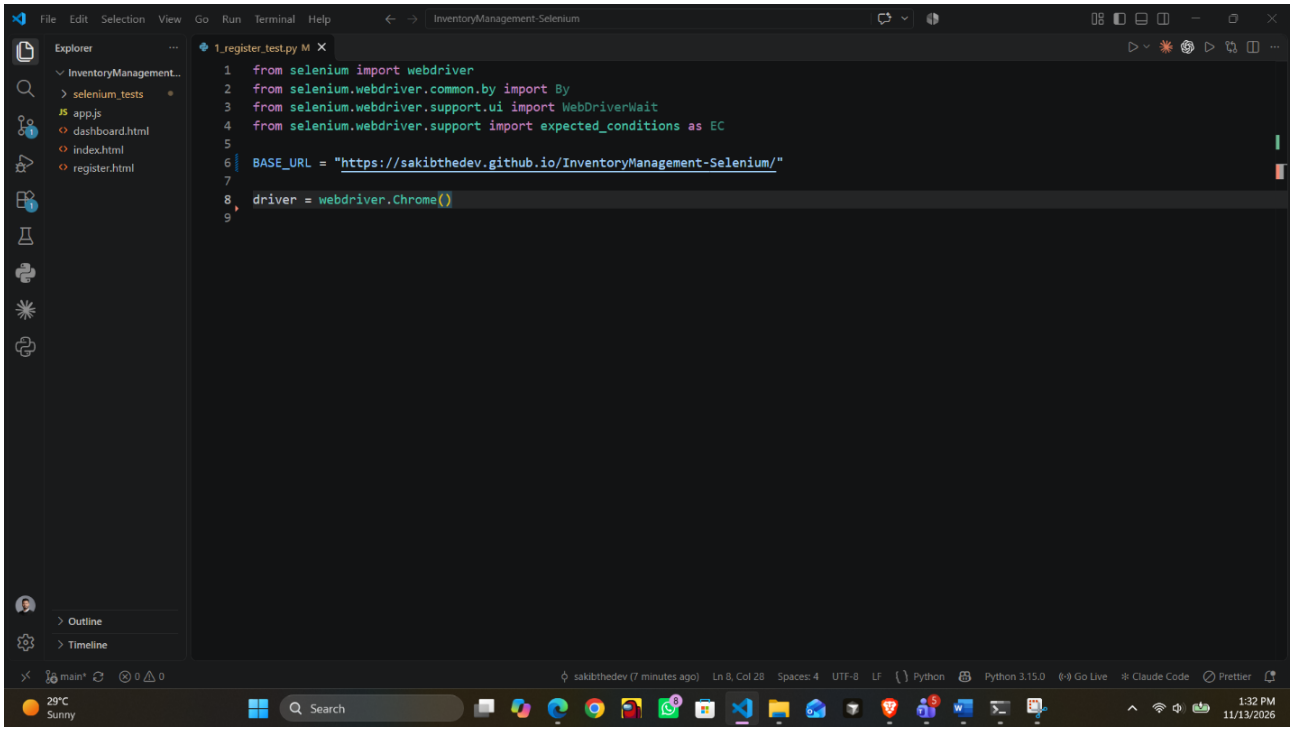


Fig. 08: Selenium imported in Visual Studio Code

3. Selenium Testing - Login

Description:

This test checks the login page of the Inventory System. The script 2_login_test.py was run for two scenarios: login with valid credentials, which should open the dashboard, and login with an invalid email, which should be rejected.

The script opens the login page and enters the email admin@test.com and the password admin123 using the element IDs login-email and login-password. It clicks the login button, waits for the URL to change to dashboard.html and asserts it. A success message is printed if the check passes, an error message is printed if it fails, and the browser is closed at the end.

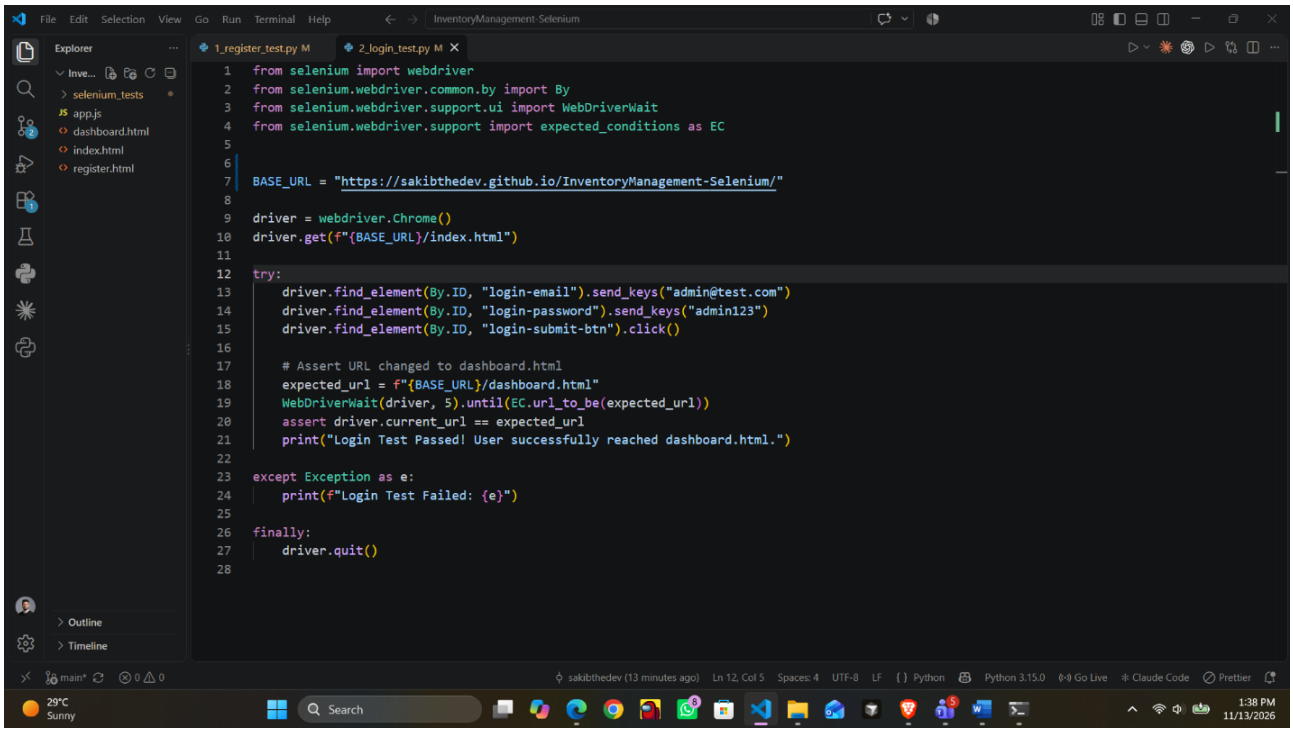


Fig. 09: Login test script

Chrome, controlled by the automated test software, opened the Inventory System login page. The valid email and password were filled in automatically before the Login button was clicked.

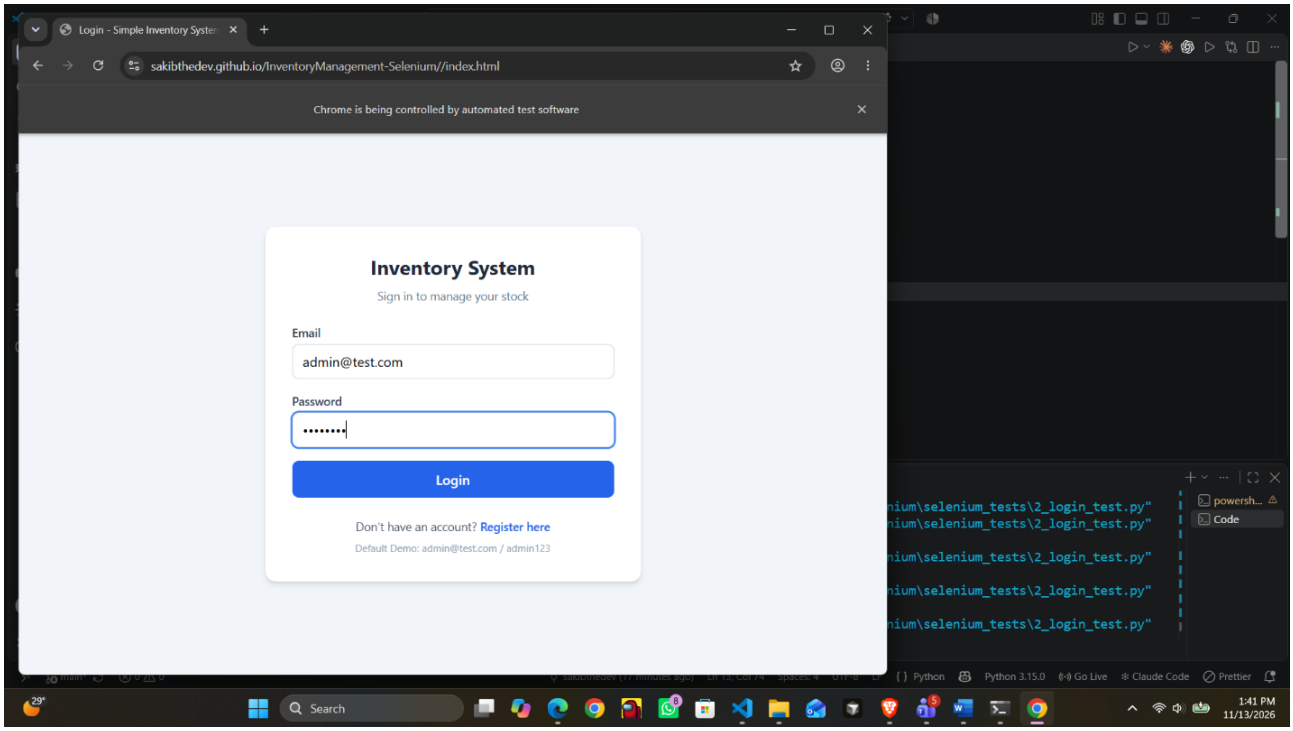


Fig. 10: Login page opened by Selenium with valid credentials

The terminal output shows the message “Login Test Passed! User successfully reached dashboard.html.”, so the valid login was successful and the user reached the dashboard.

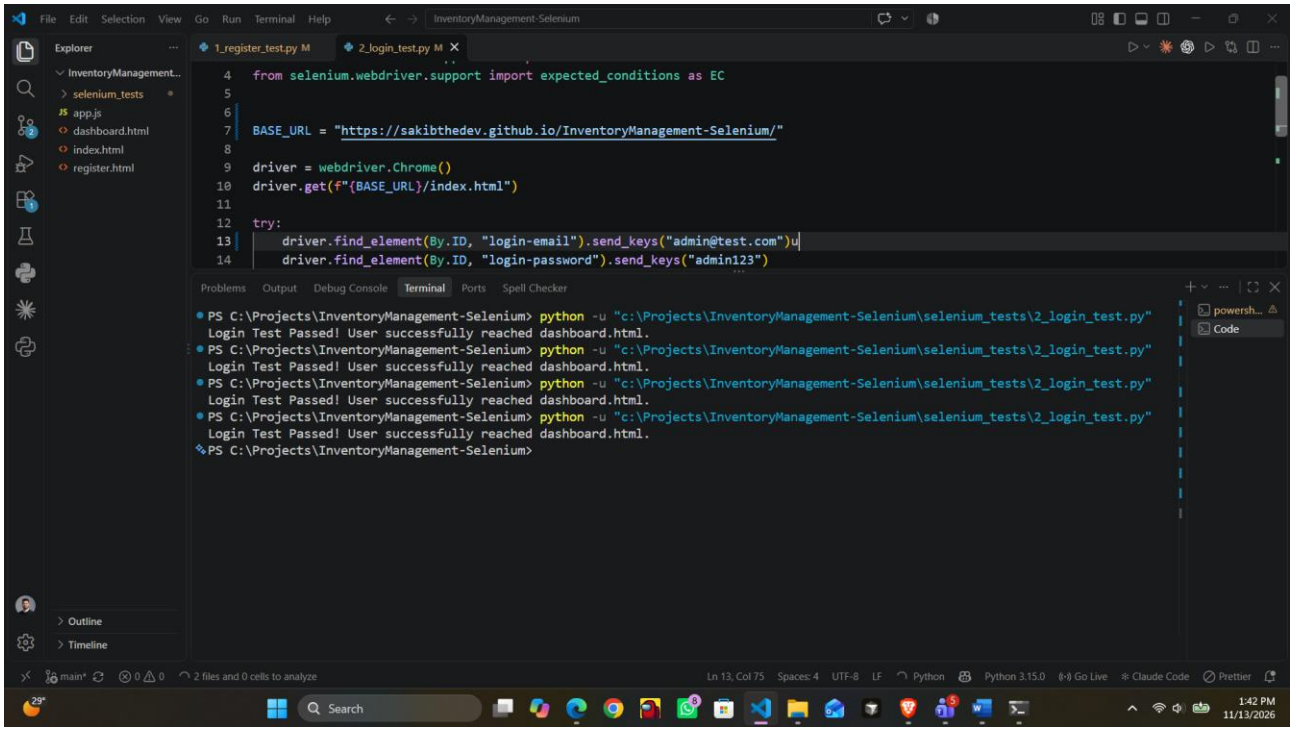


Fig. 11: Login test passed

In the negative scenario, an incorrect email (admin@test1.com) was entered. The system did not allow access, stayed on the login page and displayed the red error message “Invalid email or password!”.

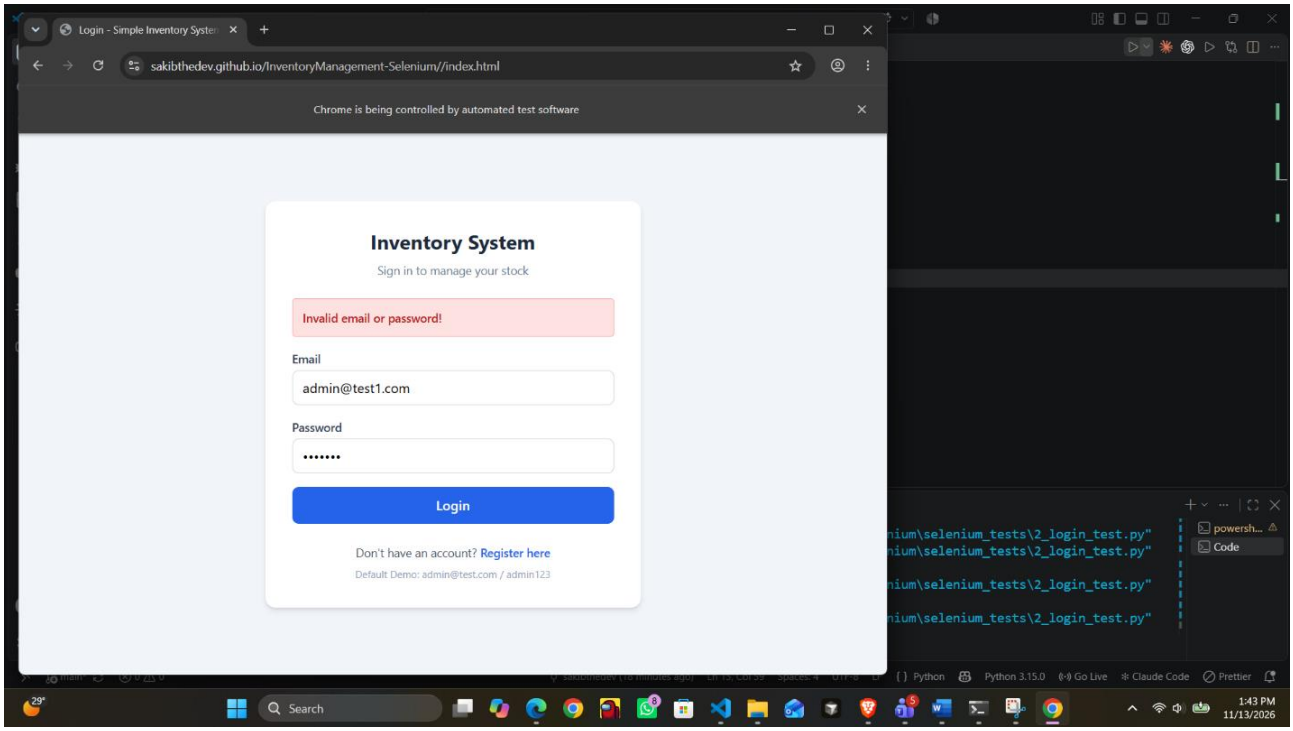


Fig. 12: Login failed with an invalid email

Test Cases - Login

Project Name: InventoryManagement-Selenium			**Test Designed by: ATIQUL ISLAM SAKIB**	
Test Case ID: TC_1			**Test Designed date:** 13/11/2026	
Test Priority (Low, Medium, High): High			**Test Executed by:** NAZMUL HAQUE NOYON	
Module Name: Login			**Test Execution date:** 14/11/2026	
Test Title: Verify login with valid email and password				
Description: Test the login page of the Inventory System				
Precondition: The user has a valid email and password				
Dependencies: None				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open the Inventory System login page. 2. Enter a valid Email. 3. Enter the correct Password. 4. Click “Login”.	Email: admin@test.com Password: admin123	System authenticates the user and opens the Inventory Dashboard (dashboard.html).	As expected	Pass

Project Name: InventoryManagement-Selenium			**Test Designed by: ATIQUL ISLAM SAKIB**	
Test Case ID: TC_2			**Test Designed date:** 13/11/2026	
Test Priority (Low, Medium, High): High			**Test Executed by:** NAZMUL HAQUE NOYON	
Module Name: Login			**Test Execution date:** 14/11/2026	
Test Title: Verify login with an invalid email				
Description: Test the login page of the Inventory System				
Precondition: The login page is open and the email used is not registered				
Dependencies: None				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open the Inventory System login page. 2. Enter an invalid Email. 3. Enter a Password. 4. Click "Login".	Email: admin@test1.com Password: admin123	System denies access, stays on the login page and displays the "Invalid email or password!" error message.	As expected	Pass

4. Selenium Testing - Registration

Description:

This test checks the registration page (register.html) of the Inventory System. The script 1_register_test.py was run for two scenarios: registering a new user, which should succeed, and registering with an email that already exists, which should be rejected.

The script opens the registration page and fills in the full name, email and password using the element IDs reg-fullname, reg-email and reg-password. It clicks the register button, waits until the URL contains index.html (the login page) and asserts it. A

success message is printed if the check passes, an error message is printed if it fails, and the browser is closed at the end.

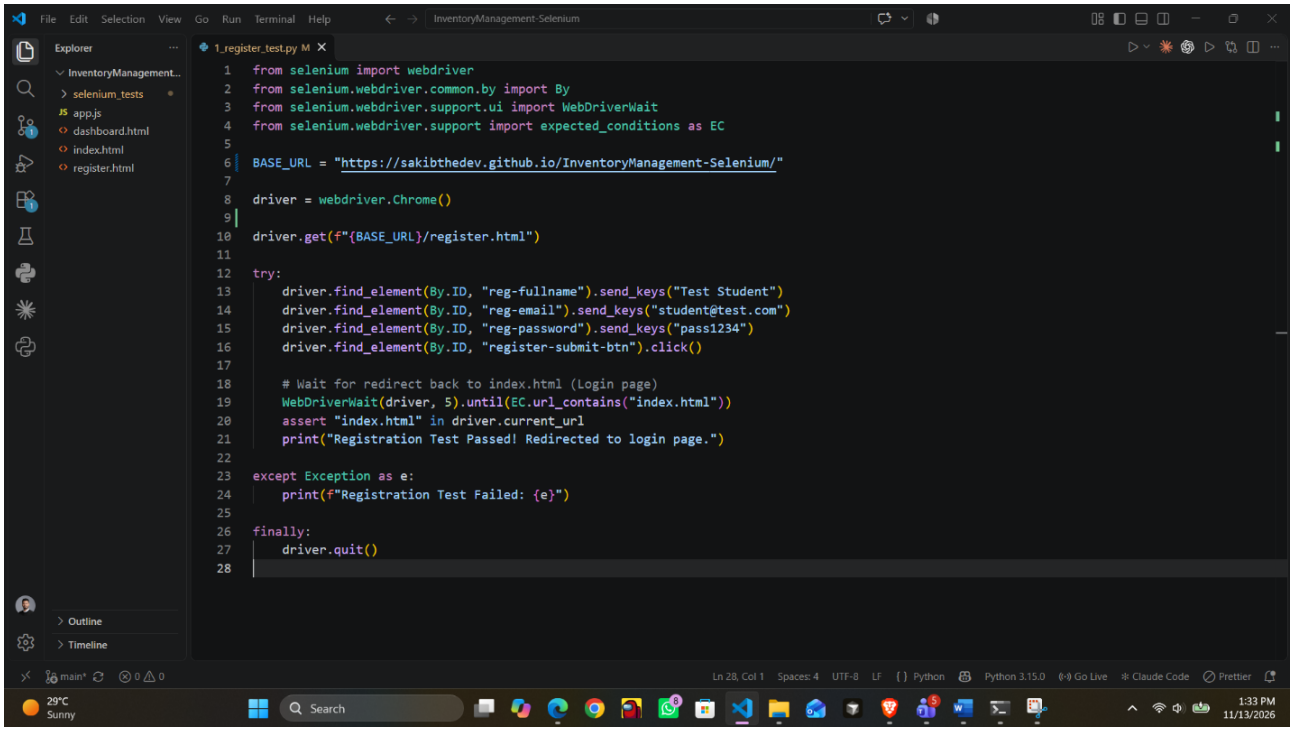


Fig. 13: Registration test script

In the successful scenario, a new user (Test Student, student@test.com) was registered. The system displayed the green message “Registration Successful! Redirecting to login...” and then moved to the login page.

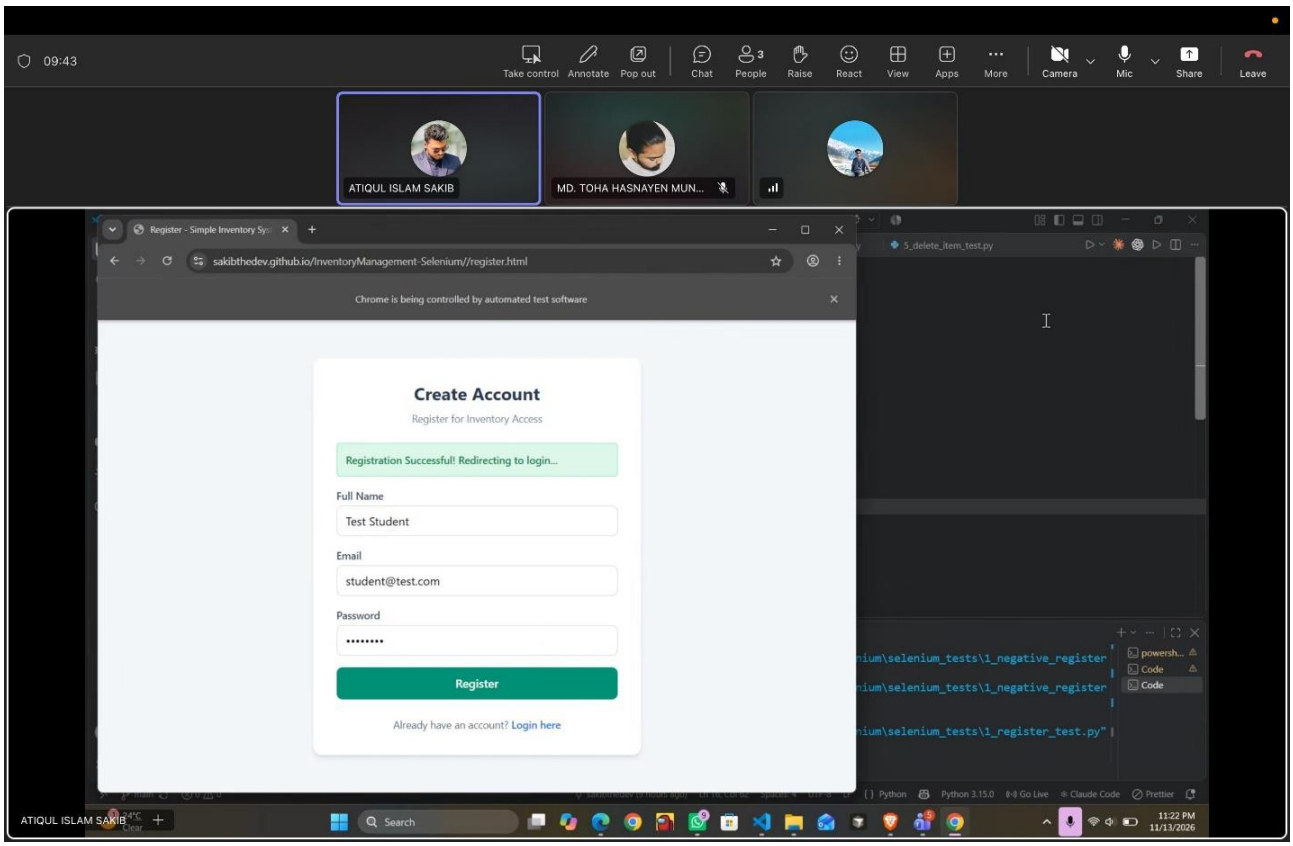


Fig. 14: Successful registration

In the failed scenario, the email admin@test.com, which is already registered, was used. The system rejected the request and displayed the red message “Email already registered!”, so no duplicate account was created.

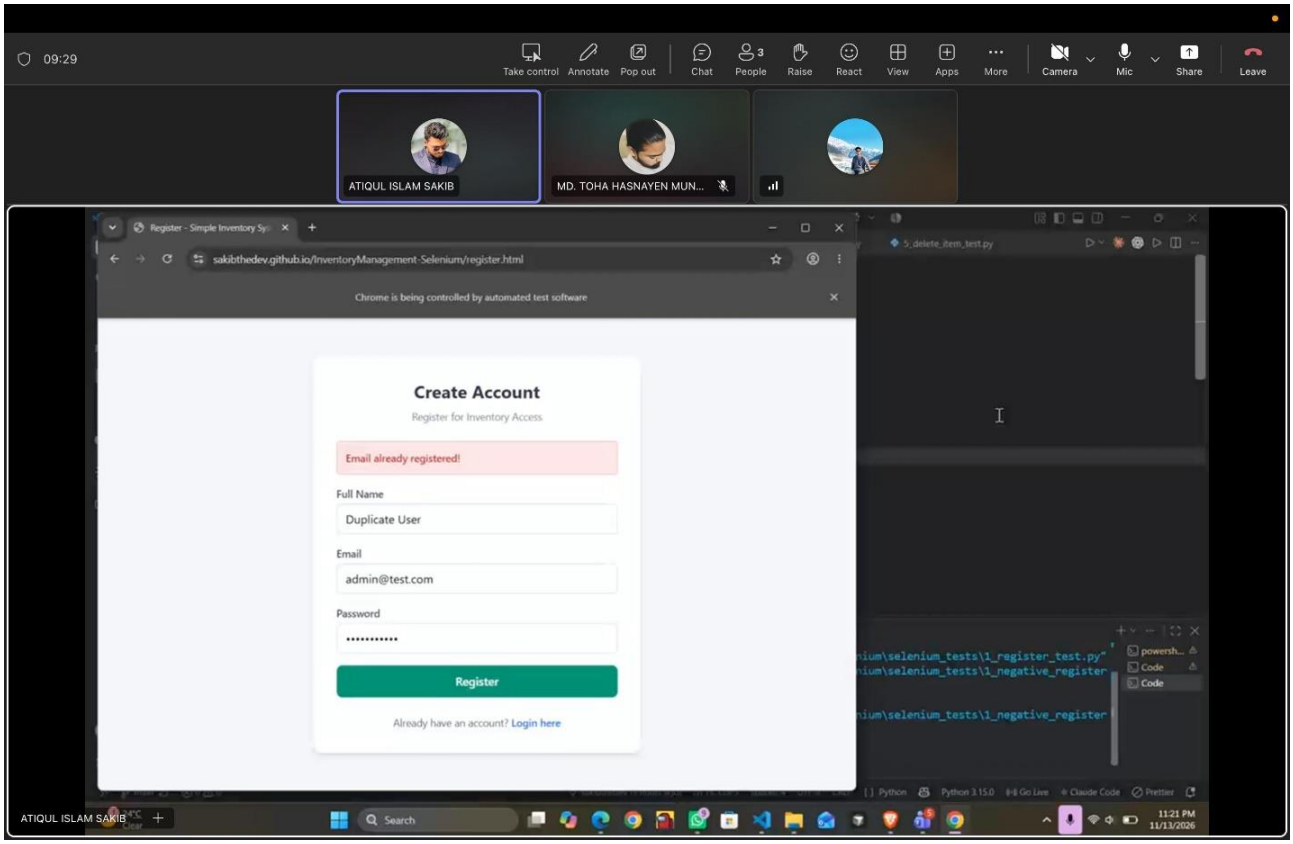


Fig. 15: Registration failed with an existing email

Test Cases - Registration

Project Name: InventoryManagement-Selenium			**Test Designed by: NAZMUL HAQUE NOYON**	
Test Case ID: TC_3			**Test Designed date:** 13/11/2026	
Test Priority (Low, Medium, High): High			**Test Executed by:** ATIQUL ISLAM SAKIB	
Module Name: Registration			**Test Execution date:** 14/11/2026	
Test Title: Verify registration with valid details				
Description: Test the registration page of the Inventory System				
Precondition: The user is not registered yet and uses an unused email				
Dependencies: None				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to the Registration screen. 2. Fill in Full Name and Email. 3. Enter a Password. 4. Click “Register”.	Name: Test Student Email: student@test.com Pass: pass1234	System creates the account, shows the “Registration Successful! Redirecting to login...” message and redirects to the Login page.	As expected	Pass

Project Name: InventoryManagement-Selenium			**Test Designed by: NAZMUL HAQUE NOYON**	
Test Case ID: TC_4			**Test Designed date:** 13/11/2026	
Test Priority (Low, Medium, High): High			**Test Executed by:** ATIQUL ISLAM SAKIB	
Module Name: Registration			**Test Execution date:** 14/11/2026	
Test Title: Verify registration with an already registered email				
Description: Test the registration page of the Inventory System				
Precondition: The email admin@test.com is already registered				
Dependencies: None				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to the Registration screen. 2. Fill in Full Name and an existing Email. 3. Enter a Password. 4. Click "Register".	Name: Duplicate User Email: admin@test.com Pass: Any valid password	System denies creating the account and displays the "Email already registered!" error message.	As expected	Pass

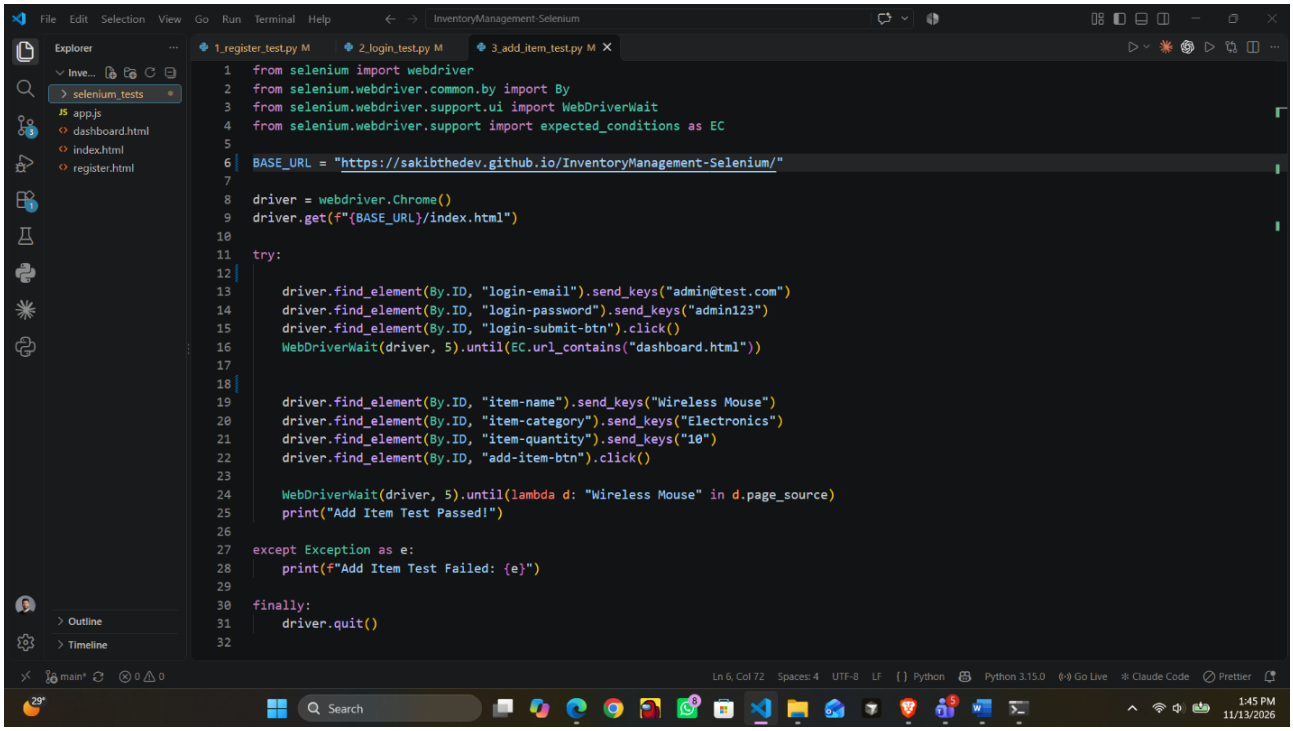
5. Selenium Testing - Add Item

Description:

This test checks the Add New Item feature of the Inventory Dashboard. The script 3_add_item_test.py logs in with the valid account and adds a new item to the stock list.

The script logs in with admin@test.com and admin123 and waits for the dashboard. It then enters the item name "Wireless Mouse", the category "Electronics" and the

quantity 10, and clicks the Add Item button. It waits until the item name appears on the page and prints a success message, or an error message if the check fails.



```python
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

BASE_URL = "https://sakibthedev.github.io/InventoryManagement-Selenium/"

driver = webdriver.Chrome()
driver.get(f"{BASE_URL}/index.html")

try:

    driver.find_element(By.ID, "login-email").send_keys("admin@test.com")
    driver.find_element(By.ID, "login-password").send_keys("admin123")
    driver.find_element(By.ID, "login-submit-btn").click()
    WebDriverWait(driver, 5).until(EC.url_contains("dashboard.html"))


    driver.find_element(By.ID, "item-name").send_keys("Wireless Mouse")
    driver.find_element(By.ID, "item-category").send_keys("Electronics")
    driver.find_element(By.ID, "item-quantity").send_keys("10")
    driver.find_element(By.ID, "add-item-btn").click()

    WebDriverWait(driver, 5).until(lambda d: "Wireless Mouse" in d.page_source)
    print("Add Item Test Passed!")

except Exception as e:
    print(f"Add Item Test Failed: {e}")

finally:
    driver.quit()
```

Fig. 16: Add Item test script

Selenium opened the Inventory Dashboard in Chrome and began filling the Add New Item form with the item name and the category. The Current Stock table shows the existing items, Laptop Stand and USB-C Hub, before the new item is added.

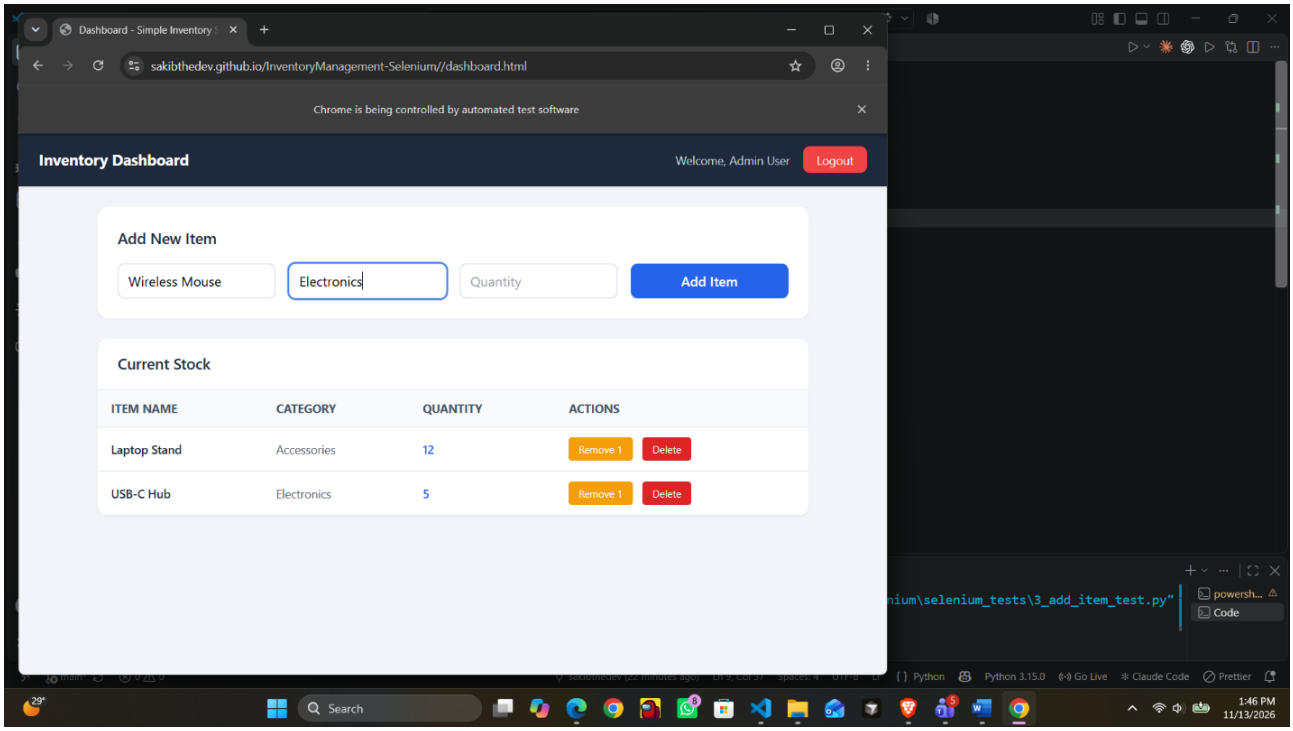


Fig. 17: Add New Item form on the dashboard

The terminal output shows the message “Add Item Test Passed!”, which confirms that the new item was added to the inventory.

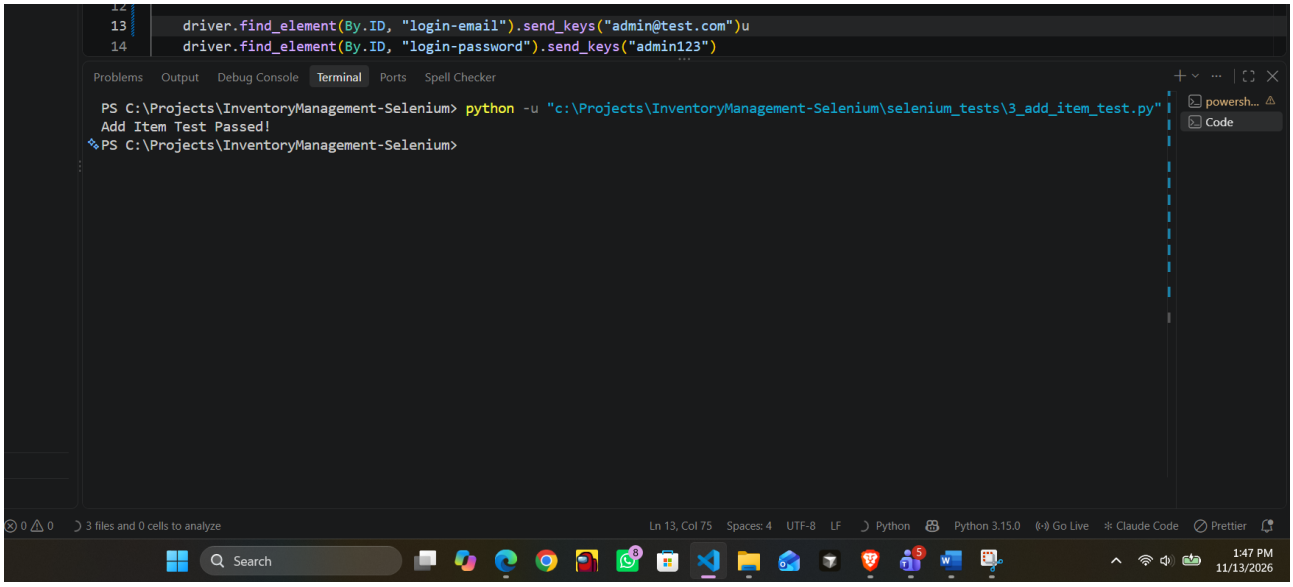


Fig. 18: Add Item test passed

Test Cases - Add Item

Project Name: InventoryManagement-Selenium			**Test Designed by: MD. TOHA HASNAYEN MUNNA**	
Test Case ID: TC_5			**Test Designed date:** 13/11/2026	
Test Priority (Low, Medium, High): High			**Test Executed by:** NAZMUL HAQUE NOYON	
Module Name: Add Item			**Test Execution date:** 14/11/2026	
Test Title: Verify adding a new item with valid details				
Description: Test the Add New Item feature of the Inventory Dashboard				
Precondition: The user is registered and the Inventory Dashboard is reachable after login (admin@test.com / admin123)				
Dependencies: Login (TC_1) must pass				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Log in with valid credentials. 2. Enter the Item Name, Category and Quantity in the Add New Item form. 3. Click “Add Item”.	Email: admin@test.com Password: admin123 Item: Wireless Mouse Category: Electronics Quantity: 10	System adds the item and shows “Wireless Mouse” in the Current Stock table.	As expected (Add Item Test Passed!)	Pass

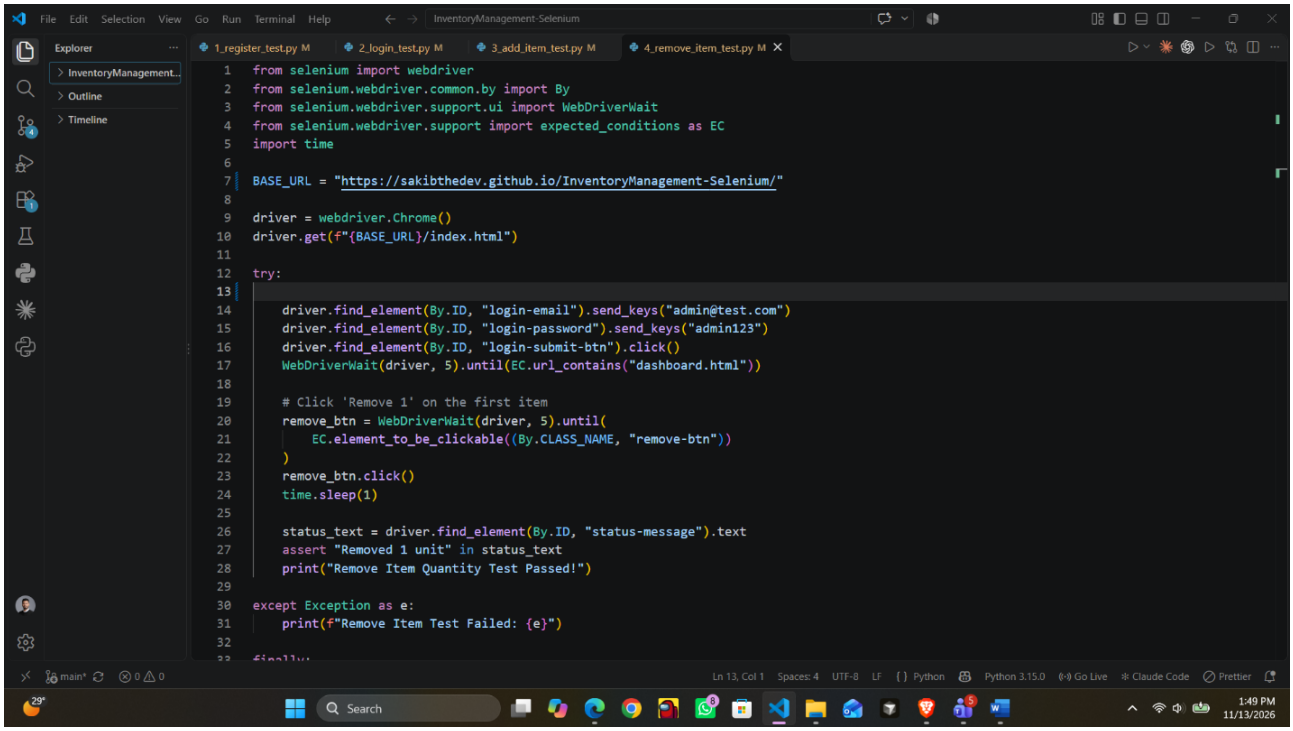
6. Selenium Testing - Remove Item

Description:

This test checks the Remove 1 button of the Inventory Dashboard, which reduces the quantity of an item by one. The script 4_remove_item_test.py logs in and removes one unit from the first item in the stock list.

The script logs in and waits for the dashboard. It waits until the first “Remove 1” button can be clicked, clicks it, and then reads the status message. It asserts that the message

contains “Removed 1 unit” and prints “Remove Item Quantity Test Passed!”, or an error message if the check fails.



```python
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

BASE_URL = "https://sakibthedev.github.io/InventoryManagement-Selenium/"

driver = webdriver.Chrome()
driver.get(f"{BASE_URL}/index.html")

try:

    driver.find_element(By.ID, "login-email").send_keys("admin@test.com")
    driver.find_element(By.ID, "login-password").send_keys("admin123")
    driver.find_element(By.ID, "login-submit-btn").click()
    WebDriverWait(driver, 5).until(EC.url_contains("dashboard.html"))

    # Click 'Remove 1' on the first item
    remove_btn = WebDriverWait(driver, 5).until(
        EC.element_to_be_clickable((By.CLASS_NAME, "remove-btn"))
    )
    remove_btn.click()
    time.sleep(1)

    status_text = driver.find_element(By.ID, "status-message").text
    assert "Removed 1 unit" in status_text
    print("Remove Item Quantity Test Passed!")

except Exception as e:
    print(f"Remove Item Test Failed: {e}")
```

Fig. 19: Remove Item test script

After the button was clicked, the green message “Removed 1 unit from item.” appeared and the quantity of Laptop Stand decreased from 12 to 11, while USB-C Hub remained at 5.

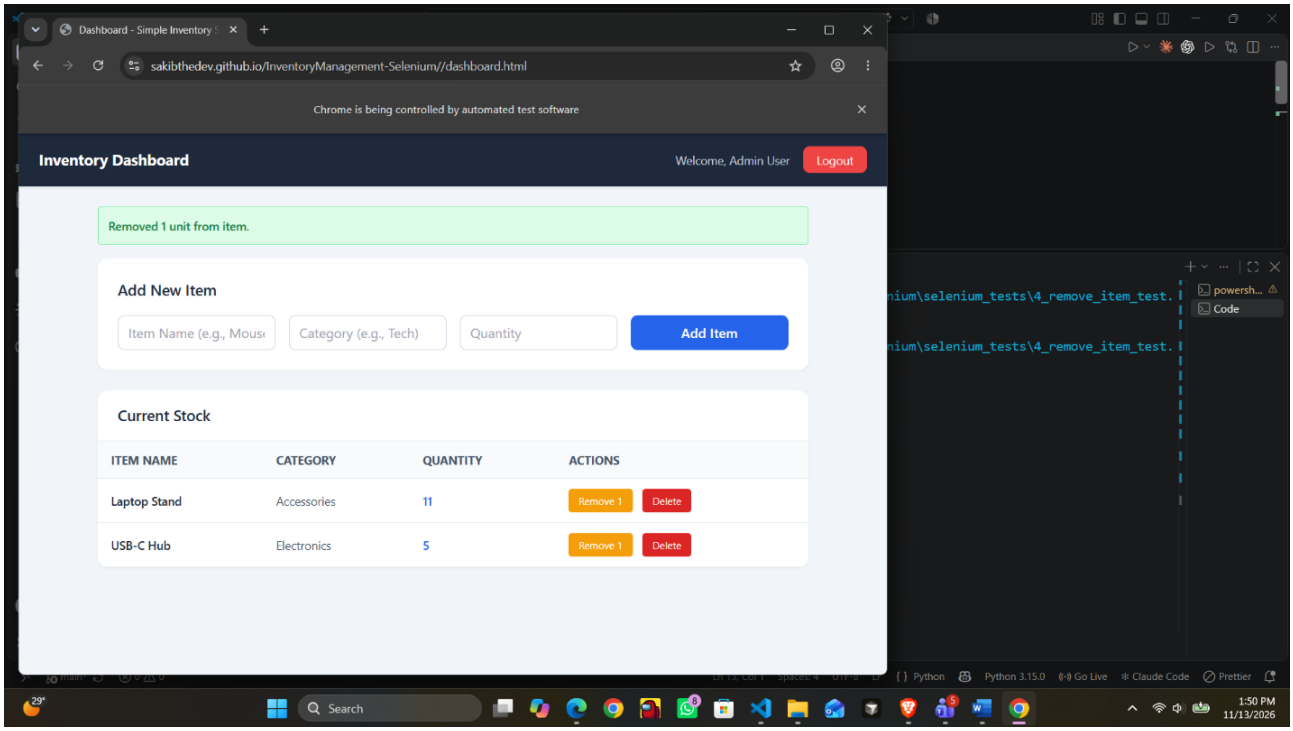


Fig. 20: Quantity reduced after clicking Remove 1

Test Cases - Remove Item

Project Name: InventoryManagement-Selenium			**Test Designed by: NAZMUL HAQUE NOYON**	
Test Case ID: TC_6			**Test Designed date:** 13/11/2026	
Test Priority (Low, Medium, High): Medium			**Test Executed by:** ATIQUL ISLAM SAKIB	
Module Name: Remove Item			**Test Execution date:** 14/11/2026	
Test Title: Verify removing one unit of an item				
Description: Test the Remove 1 button of the Inventory Dashboard				
Precondition: The user is registered and the Inventory Dashboard is reachable after login (admin@test.com / admin123); the stock list contains Laptop Stand with quantity 12				
Dependencies: Login (TC_1) must pass				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Log in with valid credentials. 2. In the Current Stock table, click "Remove 1" for the first item. 3. Check the status message and the quantity.	Email: admin@test.com Password: admin123 Item: Laptop Stand Quantity before: 12	System reduces the quantity by 1 (12 to 11) and shows the "Removed 1 unit from item." message.	As expected	Pass

7. Selenium Testing - Delete Item

Description:

This test checks the Delete button of the Inventory Dashboard, which removes an item completely from the stock list. The script 5_delete_item_test.py first adds a temporary item and then deletes it.

The script logs in and adds a temporary item named “Temp Item To Delete” with the category “Misc” and the quantity 2. It then clicks the Delete button of the last item in the list and asserts that the item name is no longer on the page. A success message is printed if the check passes.

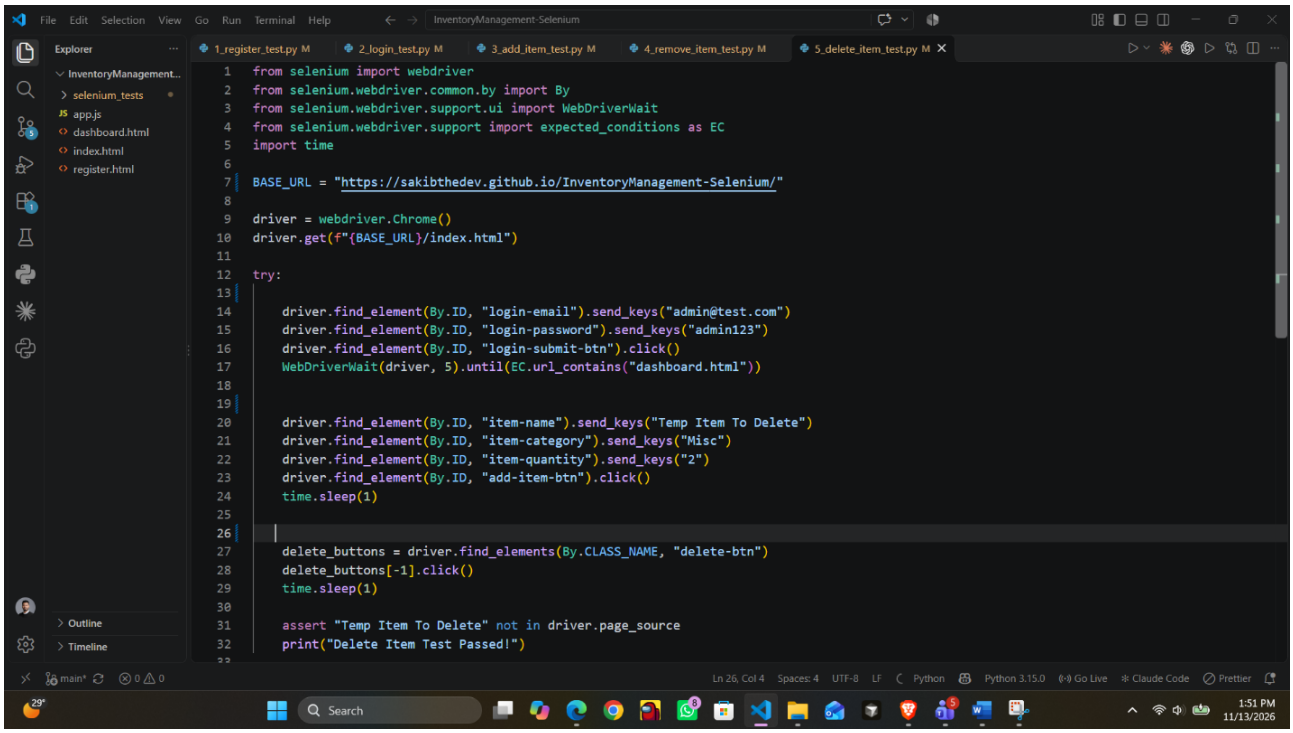


Fig. 21: Delete Item test script

The system displayed the green message “Added “Temp Item To Delete” to inventory!”, and the new item appeared as the third row of the Current Stock table with the category Misc and the quantity 2.

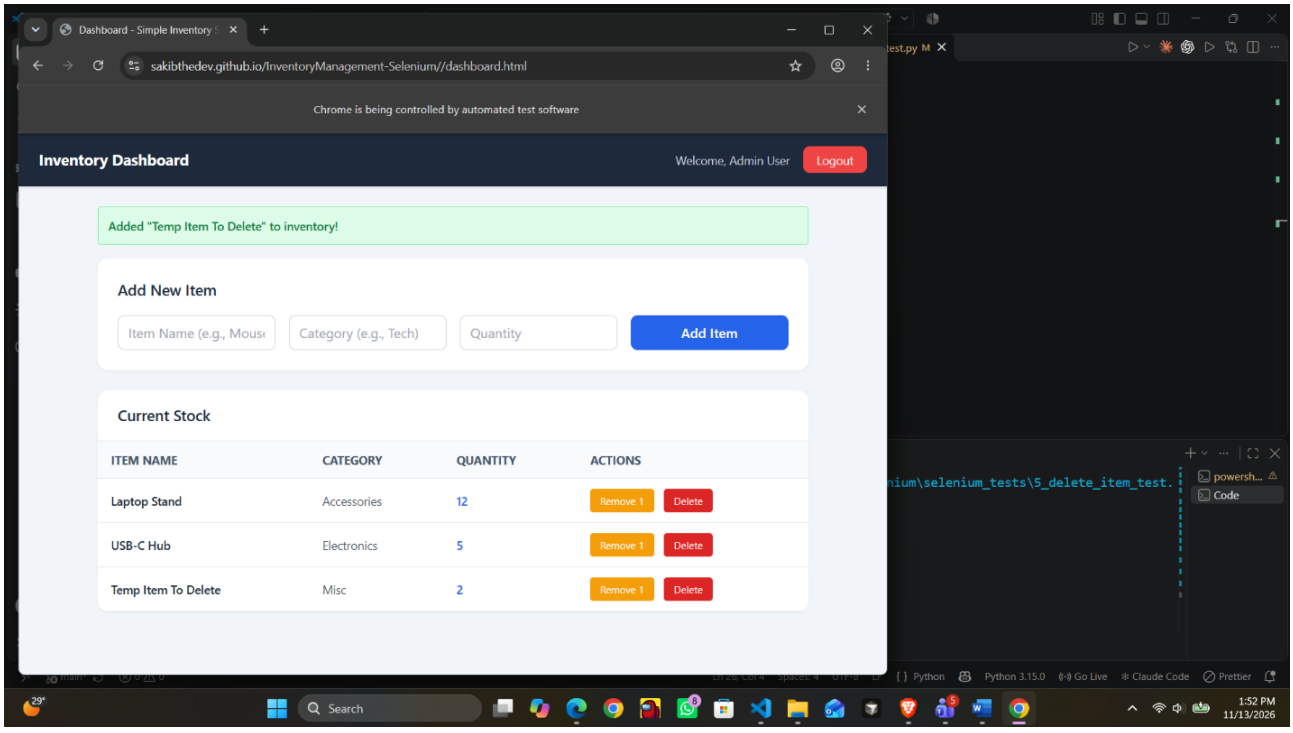


Fig. 22: Temporary item added to the inventory

After the Delete button was clicked, the system displayed the message “Item deleted from inventory.” The temporary item was removed and only Laptop Stand and USB-C Hub remained in the table.

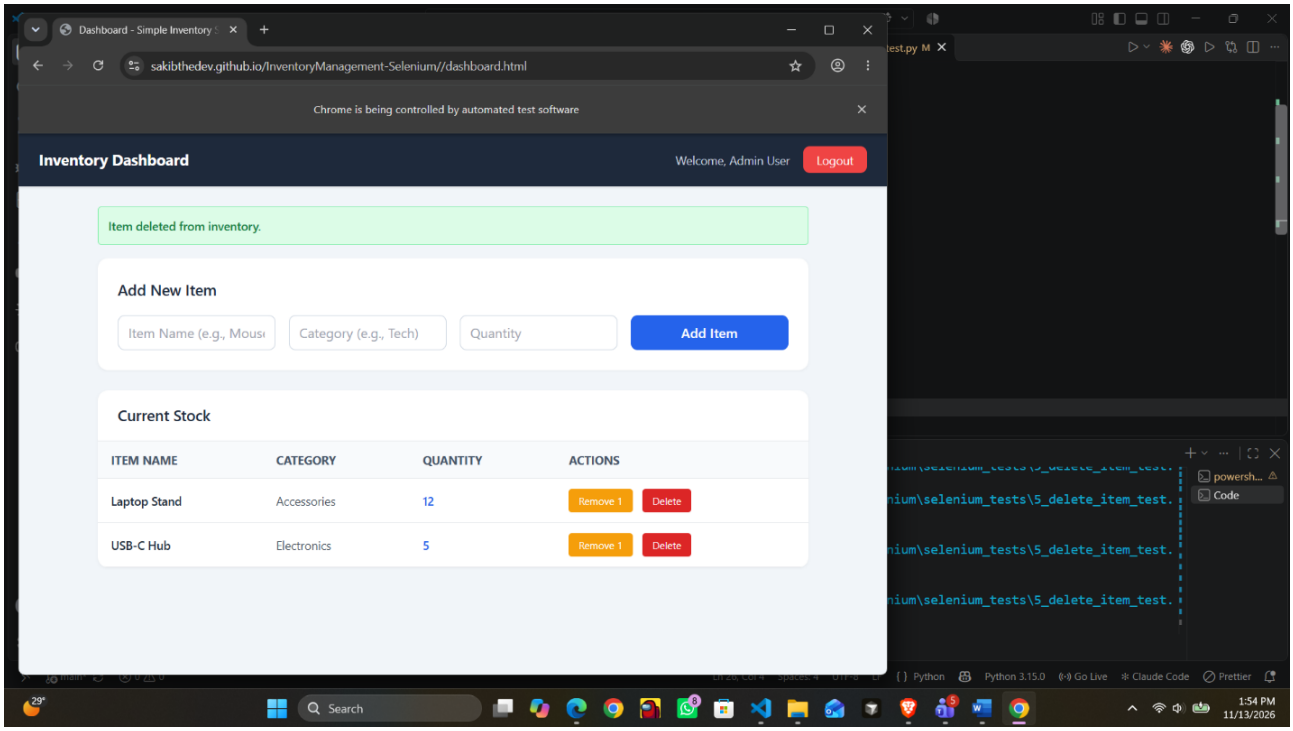


Fig. 23: Item deleted from the inventory

Test Cases - Delete Item

Project Name: InventoryManagement-Selenium			**Test Designed by: ATIQUL ISLAM SAKIB**	
Test Case ID: TC_7			**Test Designed date:** 13/11/2026	
Test Priority (Low, Medium, High): Medium			**Test Executed by:** MD. TOHA HASNAYEN MUNNA	
Module Name: Delete Item			**Test Execution date:** 14/11/2026	
Test Title: Verify deleting an item from the inventory				
Description: Test the Delete button of the Inventory Dashboard				
Precondition: The user is registered and the Inventory Dashboard is reachable after login (admin@test.com / admin123)				
Dependencies: Login (TC_1) and Add Item (TC_5) must pass				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Log in with valid credentials. 2. Add a new item using the Add New Item form. 3. Click “Delete” for the newly added item. 4. Check the status message and the stock list.	Email: admin@test.com Password: admin123 Item: Temp Item To Delete Category: Misc Quantity: 2	System deletes the item, shows the “Item deleted from inventory.” message and the item is no longer in the Current Stock table.	As expected	Pass